Christopher Hou

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Work Experience

Instrumental Palo Alto, CA
QA Engineer **Jul 2026 – Present**

- Extended a Python/TypeScript fleet-management CLI and dashboard as lead contributor (27 releases), adding a 20-minute memory-leak screen that replaces 24-hour burn-ins, one-click PR-build deploys, and remote execution
- Shipped 20+ features and fixes to the station app (TypeScript, Angular, Node.js, GraphQL), including GPU telemetry that replaced paid Datadog GPU monitoring, camera health diagnostics, and an anomaly-detection fix
- Fixed fleet-wide Linux OS bugs: an Xorg config conflict that blanked every screen after 10 idle minutes, an accessibility-bus race that crash-looped the on-screen keyboard, and oversized OTA logs that stalled error reports
- Automated monthly OS release cuts across 4 repositories with a scheduled Python/CircleCI pipeline, eliminating 6 of 8 manual steps, and led staged OTA rollouts to a 250+ device edge fleet
- Root-caused a production GPU memory-exhaustion issue that forced ML inference onto CPU, cutting inspection cycle time by over half (10+ min to 4.2 min median) and leading the postmortem
- Eliminated flaky tests across 3 repositories (TypeScript, Scala, Python); one fix resolved 94% of a month's flaky CI failures in the backend repo
- Built Playwright/TypeScript end-to-end smoke and release-candidate regression suites that validate each release on real hardware, automating 68% of the release test plan and catching memory leaks and regressions before production

ASML San Jose, CA
SQA Engineer 3 **Mar 2025 – Jul 2026**

- Developed a Python-based I/O file tracking tool adopted by 60+ engineers across 8 teams, with full documentation and iterative improvement based on user feedback
- Optimized a job-scanning diagnostic tool, reducing runtime from 17 hours to 10 minutes through algorithmic improvements
- Enhanced GPU test suite coverage 20x, cutting average bug detection time to under 1 day
- Built and maintained 25 regression suites (800+ cases), contributing 25% of PRs in the SQA repo, raised regression tracking to 100% and automation by 25%
- Led cross-team coordination across 3 product groups and 9 engineers to deliver a customer-critical feature 2 weeks ahead of schedule

ASML San Jose, CA
SQA Engineer 2 **Oct 2023 – Mar 2025**

ASML San Jose, CA
Test Engineer **Jul 2023 – Oct 2023**

Education

University of California, Berkeley Berkeley, CA
B.A. in Computer Science **December 2021**

Relevant coursework: Data Structures, Algorithms, AI, Software Engineering, Operating Systems, Databases

Project: Co-led team of 6 to improve national CS education program, boosting engagement for 600+ teachers

Projects

Warbot Suite – Discord Automation and Analytics Platform **Nov 2020 – Present**

- Architected and built a scalable Python backend with discord.py and GraphQL API integration, serving 200+ users and processing 150k+ automated messages
- Integrated LLM-powered chatbot, data visualization, and analytics tooling; deployed to AWS EC2/Oracle Cloud via PM2 with 99.9% uptime

Skills

Languages: Python, TypeScript, JavaScript, Bash, C++, Java, SQL, HTML/CSS
Tools: Git, Linux, Docker, CircleCI, Datadog, Mender (OTA updates), REST APIs, GraphQL, AWS EC2, Jira/Confluence
Testing/Automation: Playwright, Nightwatch, Robot Framework, Lua, regression automation, CI/CD, qTest
AI/ML: Claude Code, GitHub Copilot, LLM API integration, Stable Diffusion, LoRA training, voice synthesis